

Lista 03 - Expressões de Pilha

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Exercício 01

```
1 #include <stdio.h>
2 #include <ctype.h>
3 #include <string.h>
4 #include "pilha.h"
5 /*
6     Ryan
7     RA: 1840482513013
8 */
9 char *posfixa(char *e) {
10     static char s[256];
11     int j = 0;
12     Pilha P = pilha(256);
13     for(int i=0; e[i]; i++)
14         if( isdigit(e[i]) ) s[j++] = e[i];
15         else if( strchr("+-/*",e[i]) ) empilha(e[i],P);
16         else if( e[i]=='(' ) s[j++] = desempilha(P);
17         else if( e[i]==')' ) s[j++] = desempilha(P);
18     s[j] = '\0';
19     destroip(&P);
20     return s;
21 }
22
23
24 int main(void) {
25     char e[513];
26     printf("Infixa? ");
27     gets(e);
28     printf("Posfixa: %s\n", posfixa(e));
29     return 0;
30 }
31
32 }
```

main.c:28:5: warning: 'gets' is deprecated [-Wdeprecated-declarations]
28 | gets(e);
 | ^~~~~
In file included from main.c:1:
/nix/store/0zv32kh0zb4s1v4ld6mc99vmzydj9nm9-glibc-2.27-126/include/stdio.h:667:1: note: declared here
667 | extern char *gets (char *__s) __wur __attribute__((__deprecated__));
 | ^~~~~
/nix/store/7h3qngvkw6z2r8lq4j5mks4l6r5x2cq-binutils-2.34-1/libexec/gcc/x86_64-linux-gnu/9.3.0/../../../../lib64/crti.o: warning: the 'gets' function is deprecated
main.c:(.text+0x361): warning: the 'gets' function is deprecated
Infixa? 18*20+(40 - 48)*25-(13+13)
Posfixa: 18204048-251313+

Exercício 02

```

1 #include <stdio.h>
2 #include <ctype.h>
3 #include <string.h>
4 #include "pilha.h"
5 /*
6     Ryan
7     RA: 1840482513013
8 */
9 int prio(char o) {
10     switch( o ) {
11         case '(': return 0;
12         case '+':
13         case '-': return 1;
14         case '*':
15         case '/': return 2;
16     }
17     return -1;
18 }
19
20 char *posfixa(char *e) {
21     static char s[256];

```

Exercício 03

```

1 #include <stdio.h>
2 #include <ctype.h>
3 #include <string.h>
4 #include "pilha.h"
5 /*
6     Ryan
7     RA: 1840482513013
8 */
9 int prio(char o) {
10     switch( o ) {
11         case '(': return 0;
12         case '+':
13         case '-': return 1;
14         case '*':
15         case '/': return 2;
16     }
17     return -1;
18 }
19
20 char *posfixa(char *e) {
21     static char s[256];
22
23     main.c: In function 'main':
24     main.c:82:5: warning: 'gets' is deprecated
25         82 |         gets(e);
26            |         ^~~~~
27
28     In file included from main.c:1:
29     /nix/store/0zv32kh0zb4s1v4ld6mc99vmzyt...
30         667 | extern char *gets (char *__s)
31            |         ^~~~~
32
33     /nix/store/7h3gnwgvkww6z2r8lq4j5mks4l6...
34     main.c:(.text+0x60c): warning: the `g`
35     Infixa? (1+3)*3
36     Posfixa: 13+3*
37     Valor: 12

```

Exercício 04

```
1  #include <stdio.h>
2  #include <ctype.h>
3  #include <string.h>
4  #include <stdlib.h>
5  #include "pilha.h"
6
7  /*
8     Ryan
9     RA: 1840482513013
10  */
11 int prio(char o) {
12     switch(o) {
13         case '(': return 0;
14         case '+':
15         case '-': return 1;
16         case '*':
17         case '/': return 2;
18     }
19     return -1;
20 }
21
22 char *posfixa(char *e) {
```

Infixa? (25+13)*13
Posfixa: 25 13 + 13 *
Valor: 494

Exercício 05

```
1  #include <stdio.h>
2  #include <ctype.h>
3  #include <string.h>
4  #include "pilha.h"
5  /*
6     Ryan
7     RA: 1840482513013
8  */
9  char *posfixa(char *e) {
10
11     static char s[256];
12     int j = 0;
13     Pilha P = pilha(256);
14
15     for(int i=0; e[i]; i++) {
16
17         if(e[i] == ' ') continue;
18
19         if(e[i] == '(') {
20             // ignora
21         }
22     }
```

main.c: In function 'main':
main.c:77:5: warning: 'gets' is deprecated
77 | gets(e);
 | ^~~~
In file included from main.c:1:
/nix/store/0zv32kh0zb4s1v4ld6mc99vmzydj9...
667 | extern char *gets (char *__s) __
 | ^~~~
/nix/store/7h3qngvkw6z2r8lq4j5mks4l6r5x...
main.c:(.text+0x50d): warning: the `gets`
Infixa? ((V & F) | V)
Posfixa: VF&V|
Valor: 1

Exercício 06

```
1 #include <stdio.h>
2 #include <ctype.h>
3 #include <string.h>
4 #include "pilha.h"
5 /*
6     Ryan
7     RA: 1840482513013
8 */
9 int prio(char o) {
10     switch( o ) {
11         case '(': return 0;
12         case '+':
13         case '-': return 1;
14         case '*':
15         case '/': return 2;
16     }
17     return -1;
18 }
19
20 char *posfixa(char *e) {
21     static char c[256];
22
23     main.c: In function 'main':
24     main.c:119:5: warning: 'gets' is deprecated
25     119 |     gets(e);
26         |     ^~~~
27
28     In file included from main.c:1:
29     /nix/store/0zv32kh0zb4s1v4ld6mc99vmzydj9:
30     667 | extern char *gets (char *__s)
31         |     ^~~~
32
33     /nix/store/7h3qngvgkw6z2r8lq4j5mks4l6r5x:
34     main.c:(.text+0x79d): warning: the `gets'
35     Infixa? (5+3)/1-3
36     Valor: 5
37     Posfixa: 53+1/3-
38     Prefixa: -/+5313
```

Exercício 07

```
5 /*
6     Ryan
7     RA: 1840482513013
8 */
9 int prio(char o) {
10     switch( o ) {
11         case '(': return 0;
12         case '+':
13         case '-': return 1;
14         case '*':
15         case '/': return 2;
16     }
17     return -1;
18 }
19
20 char *posfixa(char *e) {
21     static char c[256];
22
23     main.c: In function 'main':
24     main.c:148:5: warning: 'gets' is deprecated
25     148 |     gets(e);
26         |     ^~~~
27
28     In file included from main.c:1:
29     /nix/store/0zv32kh0zb4s1v4ld6mc99vmzydj9:
30     667 | extern char *gets (char *__s)
31         |     ^~~~
32
33     /nix/store/7h3qngvgkw6z2r8lq4j5mks4l6r5x:
34     main.c:(.text+0x920): warning: the `gets'
35     Infixa? (1+8)*4-(2+5)/1
36     Posfixa: 18+4*25+1/-
37     Valor posfixa: 29
38     Prefixa: -*+184/+251
39     Valor prefixa: 29
```